

Juan Diego Arce Castro

Data & AI Solutions Engineer | SQL Server | RAG | Generative AI | ETL | PostgreSQL | Azure

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PROFESSIONAL SUMMARY

Systems Engineer with experience in SQL Server, data engineering, ETL, reporting, legacy modernization, and enterprise systems support within government and regulated environments. Strong background maintaining complex database-driven platforms, optimizing T-SQL processes, building Power BI reports, and supporting Azure/on-premise infrastructure. Applied AI experience includes improving RAG document analysis and debugging pipelines, OCR-based processing, PostgreSQL/vector search workflows, API-based integrations, Ubuntu Linux, Ollama/local LLM tooling, and large-scale document retrieval over 100,000+ institutional documents. Currently developing academic research focused on Retrieval-Augmented Generation, document traceability, retrieval quality, and AI response reliability.

CORE SKILLS

AI & RAG: Retrieval-Augmented Generation, document ingestion, chunking, embeddings, semantic retrieval, vector search, response traceability, reliability evaluation

Programming & APIs: Python, SQL, C#, ASP.NET MVC, REST APIs, scripting, document processing pipelines, enterprise integrations

BI & Analytics: Power BI, DAX, KPI dashboards, operational reporting, fraud/risk data analysis, executive reporting

AI Tooling: OCR, Ollama, local LLM workflows, PostgreSQL vector storage, document retrieval validation, hallucination analysis

Data Engineering: SQL Server, PostgreSQL, ETL pipelines, data extraction, data modeling, reporting workflows, audit and regulatory data processes

Databases: SQL Server 2008/2017/2019, T-SQL, stored procedures, views, triggers, query optimization, PostgreSQL, vector database concepts

Cloud & Infrastructure: Azure Virtual Machines, IIS, Azure DevOps, Active Directory, Ubuntu Linux, Windows Server environments

Enterprise Systems: Legacy modernization, production support, requirements analysis, technical documentation, stakeholder collaboration

PROFESSIONAL EXPERIENCE

INAMU - Instituto Nacional de las Mujeres, Government of Costa Rica | *Data & Systems Engineer* | 2023 - Present

- Maintain, analyze, and optimize 600+ stored procedures and 140+ tables across mission-critical institutional systems, including financial, administrative, and operational platforms.
- Develop and optimize T-SQL queries, stored procedures, views, extraction scripts, and reporting logic for audits, operational support, regulatory requests, and executive decision-making.
- Improved and debugged the document processing pipeline for an institutional RAG system, supporting ingestion, analysis, validation, and retrieval workflows for 100,000+ INAMU documents.
- Worked with API-based integrations, PostgreSQL, OCR, vector search, Ubuntu Linux, and Ollama-based AI workflows to strengthen document analysis and RAG troubleshooting capabilities.
- Support ETL and reporting workflows used for institutional, government, and audit-related data processes.
- Build and maintain Power BI dashboards and KPI reports for executive, operational, and compliance-oriented information needs.
- Support SQL Server environments across on-premise and Azure Virtual Machine infrastructure, including database troubleshooting, performance analysis, and system continuity.
- Collaborate on C#, ASP.NET MVC, REST API, and SQL-based enhancements for institutional applications and data services.
- Prepare and refine technical requirements, TDR documentation, and system specifications for public-sector technology procurement and modernization projects coordinated through SICOP.
- Work with cross-functional stakeholders to translate operational needs into technical solutions involving databases, reporting, automation, and enterprise systems.

Amazon Costa Rica - Global Services | *Fraud Risk Specialist* | Sep 2019 - May 2023

- Performed data analysis to detect anomalies and fraud patterns across global platforms and operational processes.
- Used SQL and advanced Excel reporting to support operational monitoring, investigations, and decision-making.
- Collaborated with analytics and product teams to improve internal controls and risk detection workflows.
- Built strong analytical judgment in high-volume, risk-sensitive operational environments requiring accuracy, prioritization, and process discipline.

APPLIED AI / RAG PROJECT

Retrieval-Augmented Generation System for Academic and Technical Document Queries | Academic / Applied AI Research Project | 2026

- Designed a RAG architecture to support document-based question answering over academic and technical documentation.
- Built a Python document processing pipeline for ingestion, text extraction, cleaning, chunking, vectorization, and metadata storage.
- Used PostgreSQL with vector search capabilities to store document fragments, embeddings, metadata, queries, retrieved evidence, and generated responses.
- Integrated the RAG workflow with conversational channels using OpenClaw and Discord as the user interaction layer.
- Evaluated system behavior across document traceability, retrieval quality, and response reliability.
- Defined evaluation criteria to identify hallucinations, unsupported claims, incomplete retrieval, and inconsistencies between generated answers and source documents.
- Worked with open generative models, local AI tooling, Ubuntu Linux environments, OCR/document processing, and semantic search workflows.

TECHNICAL STACK

AI & RAG: Retrieval-Augmented Generation, semantic search, embeddings, vector databases, document chunking, prompt strategies, response traceability, hallucination evaluation, OCR, Ollama, local LLM tooling

Data & Databases: SQL Server 2008/2017/2019, PostgreSQL, T-SQL, stored procedures, views, triggers, query optimization, ETL, data extraction, audit reporting

Programming: Python, SQL, C#, ASP.NET MVC, REST APIs, scripting, document processing pipelines

Cloud & Infrastructure: Azure Virtual Machines, Azure DevOps, IIS, Active Directory, Ubuntu Linux, Windows Server environments

BI & Analytics: Power BI, DAX, KPI dashboards, operational reporting, regulatory reporting, fraud/risk data analysis

EDUCATION

Bachelor's Degree - Systems Engineering | Universidad Americana | 2021

Licenciatura - Systems Engineering, emphasis in Management Informatics | Universidad Central | In Progress, expected 2026

LANGUAGES & ADDITIONAL INFORMATION

Spanish - Native | English - Advanced (C1)

Member - Costa Rican College of IT Professionals (CPIC), ID 11343 | Driver's licenses: B1 / A3